CPSC 581 – Winter 2026
Group Project 2: Expertise Sharing

Deliverable Due Date: Monday, Mar. 9th, 2026 @ 2200h

Critiques: In-lab Tuesday Mar. 10th (B01) / Thursday Mar. 12th (B02)

Changelog

V2 (2026-02-20)

- Clarified what expertise is shared.
- Clarified how video calls and screen sharing may be used.
- Clarified the expected level of fidelity for the system implementation.

Sketch, design, implement, and document a desktop/mobile web/software system that helps experts with 3D computer aided design (CAD) software (e.g., Blender, Autodesk Fusion) **synchronously and remotely share that expertise** with novice users. The system should make creative use of *embodied input* (e.g., using sensing, tracking) and *output* (e.g., visuals, audio, haptics).

You define:

- What device type(s) the system works on (e.g., laptop, smartphone, tablet), keeping in mind that different devices have different input hardware
- What expertise is being conveyed
- How the expertise is conveyed and received
- How many experts and novices use the system simultaneously

The main evaluation criteria for a successful system are:

- Novel use of *embodied input* and *output* to convey and receive expertise
  - *Note: Video calls and screen sharing are the two most obvious ideas that come to mind. Designs that rely on these as the core interactions may suffer in this evaluation criterion. If you choose to use these interactions, consider creative ways they could be incorporated.*
- Successfully assisting the expert with conveying and the novice with receiving feedback
  - *Note: "Generalizability" to contexts beyond CAD is **not** a criterion that will be assessed. Designs that prioritize generalizability over other key design criteria may suffer in this evaluation criterion.*

Deliverables

- **Demonstration video**: A video, max. 2 minutes, showing the main functionality of the system. Can include live footage with people using the system (does not have to be the screen only). Can be annotated with text/voice overlays to clarify functionality; do not include background music. Aesthetic appeal of the video itself is not considered as long as it's clear.
- **Write-up**: A document in PDF format describing the system design process. Must include:
    - A description of the design process taken to arrive at the system design. Include details such as justification for design decisions, and the methods and tools used to diverge, converge, evaluate, compare, benchmark, etc. candidate design elements.
    - Figures (screenshots/photos/etc.) of all design process artifacts that informed the system design. All figures should include brief descriptions/annotations, to the extent that someone from outside your team could understand what you did. Examples of artifacts may include (but not limited to):
        - Sketches;
        - Prototypes;
        - Results from individual and team ideation activities;
        - Results from comparisons/evaluations/benchmarking of alternative design candidates (e.g., through prototype testing, comparison tools).
    - A list of any functionality that is known to be incomplete, broken, or missing at the time of submission.
- **Source code**: A link to a public repository or zip file containing the code for the system.
- **The system:** A link to or copy of a *build* of the system with instructions to run it. The system must work well enough to demo.
    - Note: A more functional system will result in better grades for system completeness & reliability and the demo. Some functionality (e.g., real-time video) may be challenging to implement in the time frame of the project. The grading will account for this, but do your best to complete challenging aspects. High-fidelity prototyping of system components may be important to test suitability of design alternatives against the design criteria.

Critique ("Crit")

Critique will be held in-lab the week of the project due date. Details about the critique format:

- There will be 10 minutes of time at the start of the lab to set up your team's station.
- The course instructor and TA will independently visit each team for a presentation. Each visit duration is 10 minutes, in which the presentation portion is 3 minutes, followed by a 7-minute Q&A period. *Have documentation ready during the Q&A to assist in answering specific questions.*

- All members of your team must remain present in the lab until evaluated by both the instructor and the TA.
- While not being evaluated, teams may visit each other's stations.

Grading

By default, team members will all receive the same grade. However, if team feedback survey reviews indicate that members of the team did not contribute as fully as others, they may be given a lower grade for the project. In these cases, the instructor will discuss the contribution discrepancy with the team to help determine the grade for each member of the group.

Late Work

Unless alternative arrangements have been made, late submission materials will be graded as follows:

- After the deadline up to 24 hours late: -25%
- After 24 hours up to 48 hours late: -50%
- Over 48 hours late: -100% (i.e., not accepted)

Team members absent from critique will receive a critique grade of 0 for that critique, unless alternative arrangements have been made.

Use of Generative AI (GenAI)

Use of GenAI tools is acceptable, but should be cited and credited appropriately in written and/or presentation materials: Clearly state what tools you used, how they were used (e.g., your prompts), and how you included this into your work. Failure to cite the use of AI-generated content in the project will be considered a breach of academic integrity and subject to Academic Misconduct procedures. Students are ultimately accountable for the work they submit.